TrendPilot: A Local Agentic AI Assistant Using Ollama

AIRI Team PITB - AI Internship Task 2
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1. Project Overview

TrendPilot is a local agentic AI assistant built with Python and Ollama using the Gemma 3:4B model. Given a topic, platform, tone and output type, it plans its work, calls eight tools, reviews its own output, rewrites the caption if needed, and saves the final result as a Markdown file. It solves the problem of turning a raw project idea into a platform-ready content plan (idea, hook, caption, hashtags, reel script, title and thumbnail text) without any cloud API.

2. Tools and Technologies

Python, Ollama, Gemma 3:4B, requests, JSON memory, Markdown file saving, FastAPI, Streamlit, CLI, ReportLab, GitHub.

3. Agent Workflow

User input (topic, platform, tone, output type) → Agent builds a plan → Trend Idea Generator creates 5 angles → Agent picks the best angle and audience → Caption Writer → Hashtag Generator → Reel Script Generator → Title/Thumbnail Generator → Content Reviewer → (if verdict is IMPROVE or score < 7) Caption Writer rewrites using the reviewer's feedback → File Saver → Memory Tool → final output. Every tool call is printed as [Tool Called]: name.

4. Tools Implemented

Tool	What it does
Trend Idea Generator	Produces 5 specific content angles; uses memory to avoid repeating old angles.
Caption Writer	Writes the hook and caption following platform rules; also used for the improvement rewrite.
Hashtag Generator	Creates relevant hashtags, de-duplicated and limited per platform.
Reel Script Generator	Writes a 4-scene, ~30 second script with visuals and voiceover.
Title & Thumbnail Generator	Creates a catchy title and short thumbnail text.
Content Reviewer	Scores the content 1-10 and checks hook, clarity, length, tone, hashtags.
File Saver	Saves the final result to .md (or .json) in outputs/.
Memory Tool	Stores previous topics, platforms, tones and saved files in memory.json.

5. Demo and Outputs

Demos: CLI (python demo.py), Streamlit (streamlit run app/streamlit_app.py) and FastAPI (uvicorn app.main:app, POST /generate-content). Generated files are in outputs/.

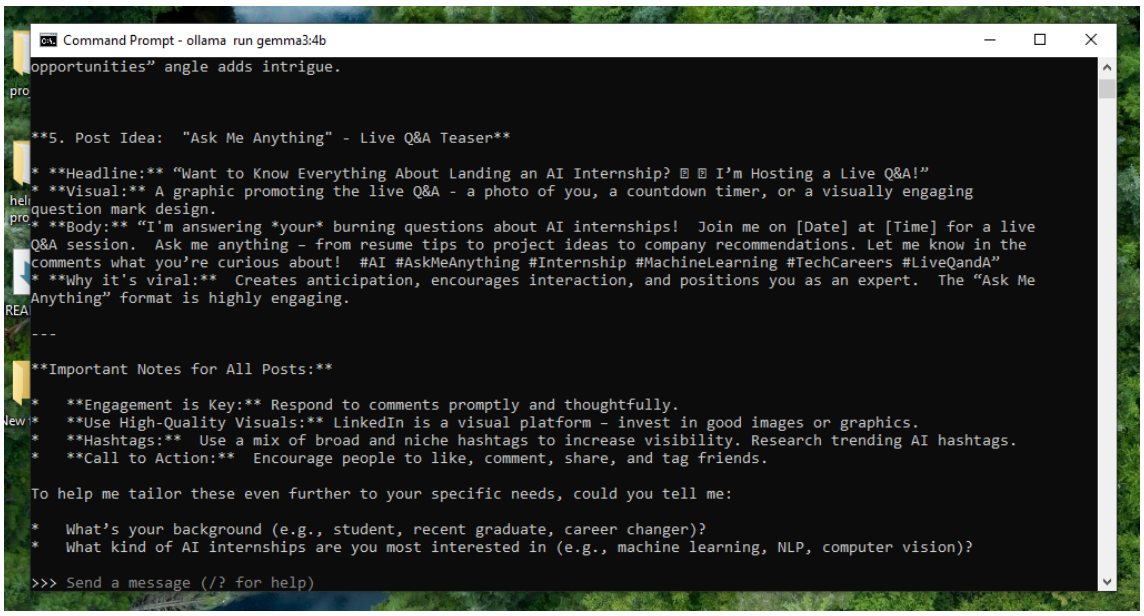


Figure 1: Ollama running the local model

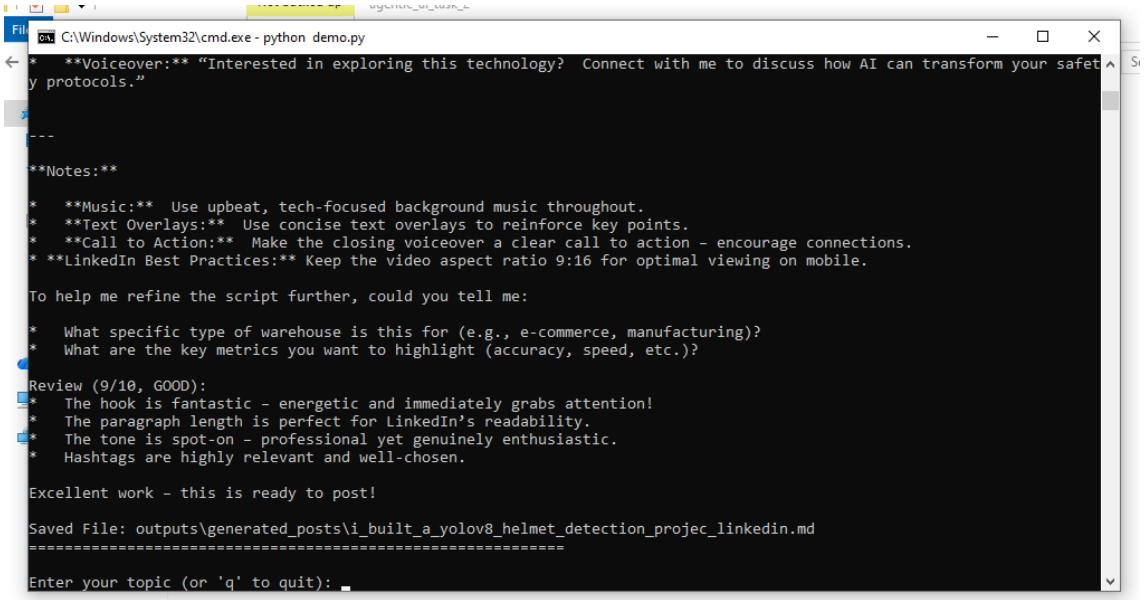


Figure 2: Agent output in the demo

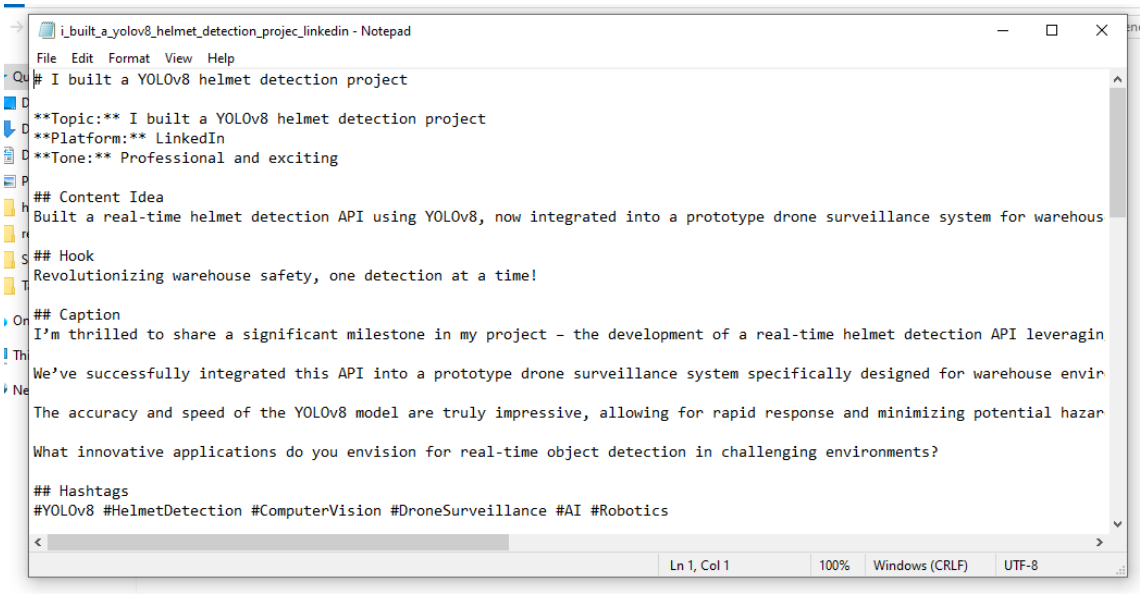


Figure 3: Saved Markdown output file

6. Testing (10 test cases)

#	Topic	Platform	Tone	Tools used	Score	Saved file	Useful?
1	YOLOv8 helmet detection	LinkedIn	Professional	Trend Idea Generator, Caption Writer, Hashtag Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8	outputs\generated_posts\yolov8_helmet_detection_linkedin.md	Yes
2	AI internship experience	LinkedIn	Motivational	Trend Idea Generator, Caption Writer, Hashtag Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8	outputs\generated_posts\ai_internship_experience_linkedin.md	Yes
3	Python learning journey	Instagram	Funny	Trend Idea Generator, Caption Writer, Reel Script Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8	outputs\generated_scripts\python_learning_journey_instagram.md	Yes
4	Machine learning project	X/Twitter	Short and catchy	Trend Idea Generator, Caption Writer, Hashtag Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8	outputs\generated_posts\machine_learning_project_x_twitter.md	Yes
5	Data annotation struggles	LinkedIn	Humorous	Trend Idea Generator, Caption Writer, Hashtag Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8	outputs\generated_posts\data_annotation_struggles_linkedin.md	Yes
6	Model failed in production	Instagram	Meme style	Trend Idea Generator, Caption Writer, Reel Script Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	6	outputs\generated_scripts\model_failed_in_production_instagram.md	No - Humorous tone requested but reviewer scored it 6/10 / asked for rewrit
7	Final year project	LinkedIn	Professional	Trend Idea Generator, Caption Writer, Hashtag Generator, Reel Script Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8	outputs\generated_posts\final_year_project_linkedin.md	Yes
8	Computer vision project	YouTube Shorts	Exciting	Trend Idea Generator, Caption Writer, Reel Script Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8	outputs\generated_scripts\computer_vision_project_youtube_shor.md	Yes
9	Debugging FastAPI	X/Twitter	Funny	Trend Idea Generator, Caption Writer, Hashtag Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8	outputs\generated_posts\debugging_fastapi_x_twitter.md	Yes
10	Open-source contribution	LinkedIn	Inspirational	Trend Idea Generator, Caption Writer, Hashtag Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	9	outputs\generated_posts\open_source_contribution_linkedin.md	Yes

Automatic quality checks (length, hashtags, format, tone): 9/10 passed in the first run, 9/10 passed after the prompt fixes.

7. Error Analysis

Test case	Problem found	Possible reason	Fix applied
Machine learning project (X/Twitter, Short and catchy)	Output was generic on the first draft: Reviewer flagged the first draft (score 8/10): * Hook could be tighter – “AI Kicks!” is punchier. * “Grail kicks” is a bit verbose; simplify. * Consider a stronger call to action. #	Prompt did not ask for specific details	Asked for specific details in angles and relied on the reviewer rewrite loop
Debugging FastAPI (X/Twitter, Funny)	Humor / meme style not convincing: Humorous tone requested but reviewer scored it 8/10 / asked for rewrite	Model did not understand meme style from a tone word alone	Added a humor instruction with an example to the caption and reel prompts
Model failed in production (Instagram, Meme style)	Output was generic on the first draft: lowest reviewer score 7/10	Prompt did not ask for specific details	Asked for specific details in angles and relied on the reviewer rewrite loop
YOLOv8 helmet detection (LinkedIn, Professional)	Output was generic on the first draft: lowest reviewer score 8/10	Prompt did not ask for specific details	Asked for specific details in angles and relied on the reviewer rewrite loop
AI internship experience (LinkedIn, Motivational)	Output was generic on the first draft: lowest reviewer score 8/10	Prompt did not ask for specific details	Asked for specific details in angles and relied on the reviewer rewrite loop

Suggested improvements

1) Add few-shot examples to prompts (especially humour / meme style). 2) Add validation checks (word count, character limit for X, hashtag count). 3) Add a user-feedback loop and a web-search tool. 4) Try a larger local model.

8. What I Learned

- How to run local LLMs with Ollama
- Difference between a chatbot and an agent
- Connecting tools to an LLM
- Prompt design for structured output
- Saving results and basic memory
- Testing and improving an agentic workflow

9. Future Improvements

Web search, multi-agent workflow (planner / writer / reviewer), voice input, PDF export, SQLite memory, user feedback loop.